# THE DIMENSION FUNCTION OF THE NORMALISED DUALIZING COMPLEX DIFFERS FROM THE BASE ONE BY THE TRANSCENDENCE DEGREE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0AWM (sources/stacks-duality/pinned-original-source.tex line 5856).

*Allowed standard prerequisites (editorial).*

- The dimension function attached to a dualizing complex and universal catenarity. Same chapter, Lemma labelled lemma-dimension-function-scheme, cited at proof lines 5908-5910 and in the statement. Supplies the dimension functions and the catenarity of S.
- Dimension functions propagate and are unique up to a locally constant difference. Morphisms, Lemma labelled morphisms-lemma-dimension-function-propagates, cited at proof lines 5904-5908, and Topology, Lemma labelled topology-lemma-dimension-function-unique, cited at proof lines 5910-5912. Reduce the claim to a single point.
- Existence of a closed point and normalisation at closed points. Properties, Lemma labelled properties-lemma-locally-Noetherian-closed-point, cited at proof lines 5914-5916, and Same chapter, Lemma labelled lemma-good-dualizing-normalized, cited at proof lines 5916. Gives the point where the equality is checked.

*Source reference.* Original lemma 0AWF: sources/stacks-duality/pinned-original-source.tex, source lines 567–567. Same-chapter pointer into the author environment lemma at source line 567; invoked at proof line 5881 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 02JW: sources/stacks-chow/tags.txt, source lines 3351–3351. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5879 of the counted proof. It is

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*Source reference.* Original lemma 02IB: sources/stacks-chow/tags.txt, source lines 3294–3294. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5882 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 02IL: sources/stacks-chow/tags.txt, source lines 3304–3304. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5885 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0AWL: sources/stacks-duality/pinned-original-source.tex, source lines 5805–5805. Same-chapter pointer into the author environment lemma at source line 5805; invoked at proof line 5888 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $S$ be a Noetherian scheme and let $\omega_S^\bullet$ be a dualizing complex. Let $f : X \to S$ be of finite type and let $\omega_X^\bullet$ be the dualizing complex normalized relative to $\omega_S^\bullet$. For all $x \in X$ we have*

$$\delta_X(x) - \delta_S(f(x)) = trdeg_{\kappa(f(x))}(\kappa(x))$$

*where $\delta_S$, resp. $\delta_X$ is the dimension function of $\omega_S^\bullet$, resp. $\omega_X^\bullet$, see Lemma 0AWF.*

*Proof.* We may replace $X$ by an affine neighbourhood of $x$. Hence we may and do assume there is a compactification $X \subset \overline{X}$ over $S$. Then we may replace $X$ by $\overline{X}$ and assume that $X$ is proper over $S$. We may also assume $X$ is connected by replacing $X$ by the connected component of $X$ containing $x$. Next, recall that both $\delta_X$ and the function $x \mapsto \delta_S(f(x)) + \mathrm{trdeg}_{\kappa(f(x))}(\kappa(x))$ are dimension functions on $X$, see Morphisms, Lemma 02JW (and the fact that $S$ is universally catenary by Lemma 0AWF). By Topology, Lemma 02IB we see that the difference is locally constant, hence constant as $X$ is connected. Thus it suffices to prove equality in any point of $X$. By Properties, Lemma 02IL the scheme $X$ has a closed point $x$. Since $X \to S$ is proper the image $s$ of $x$ is closed in $S$. Thus we may apply Lemma 0AWL to conclude. □